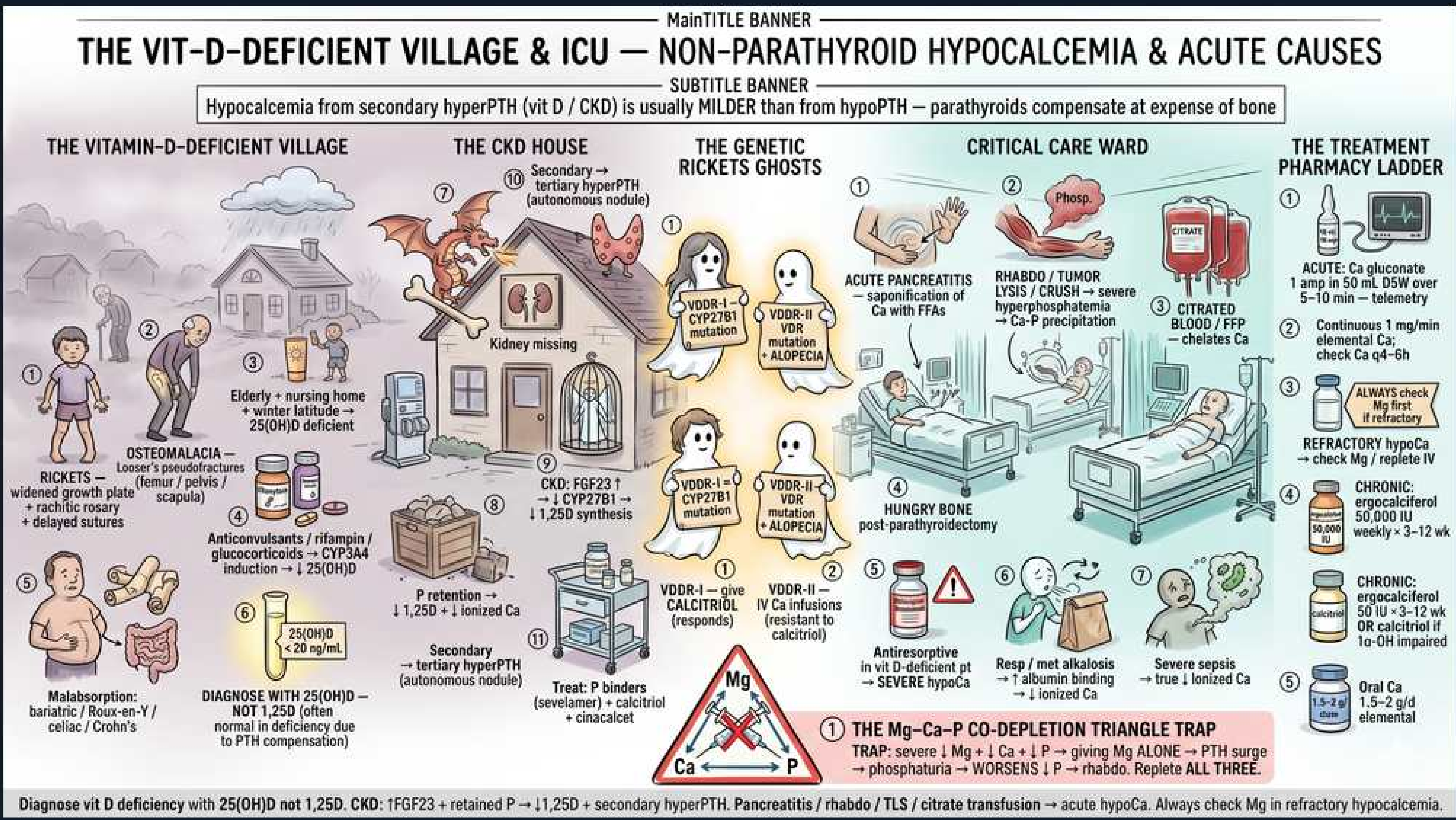
Non-Parathyroid Hypocalcemia — Vitamin D, CKD, Acute & Genetic Causes



1. Rickets / osteomalacia (vit-D deficient)

WHAT YOU SEE
Bowed-legs child, village

WHAT IT ENCODES
Vitamin D deficiency → impaired gut Ca/PO4 absorption → hypocalcemia → secondary hyperPTH. Children: rickets (widened growth plates, bowing). Adults: osteomalacia (Looser pseudofractures, hip/scapula).

BOARD TRAP
Non-parathyroid (secondary hyperPTH) hypocalcemia is usually MILDER than true hypoPTH because intact parathyroids compensate — at the expense of bone (high ALP, high PTH).

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2. Diagnose with 25(OH)D — NOT 1,25

WHAT YOU SEE
Lab vial marked 25-OHD

WHAT IT ENCODES
To diagnose vitamin D deficiency, measure 25(OH)D (storage form), NOT 1,25(OH)2D. In deficiency, 1,25 may be normal/high from compensatory secondary hyperPTH.

BOARD TRAP
Ordering 1,25-D to screen for deficiency is the classic error — it can be falsely reassuring; 25-OHD is the correct status test.

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3. Malabsorption — celiac/Crohn/bariatric

WHAT YOU SEE
Gut/Whipple figure

WHAT IT ENCODES
Fat malabsorption (celiac, Crohn, pancreatic insufficiency, Roux-en-Y bypass) wastes fat-soluble vitamin D → deficiency, secondary hyperPTH, osteomalacia.

BOARD TRAP
Post-bariatric patients need lifelong Ca + vitamin D; new bone pain or fractures years later = malabsorptive osteomalacia, not osteoporosis.

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4. Anticonvulsants ↑ CYP3A4

WHAT YOU SEE
Pill bottle accelerating catabolism

WHAT IT ENCODES
Phenytoin, phenobarbital, rifampin, and glucocorticoids induce CYP3A4 → accelerate 25(OH)D catabolism → drug-induced osteomalacia/hypocalcemia.

BOARD TRAP
A patient on chronic antiepileptics with low Ca and high ALP has drug-induced vitamin D catabolism — supplement; don't just blame diet.

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5. CKD: ↑FGF23 → ↓1,25-D → 2° hyperPTH

WHAT YOU SEE
CKD haunted house, kidney

WHAT IT ENCODES
CKD-MBD: phosphate retention raises FGF23 then PTH; loss of renal 1-alpha-hydroxylase lowers calcitriol → hypocalcemia → secondary (then tertiary) hyperPTH. Sequence: high FGF23 first, then low 1,25, then high PTH.

BOARD TRAP
Tertiary hyperPTH = autonomous nodular gland with HYPERcalcemia after long secondary disease — opposite Ca direction; may need parathyroidectomy/cinacalcet.

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6. CKD treatment: phosphate binders + cinacalcet

WHAT YOU SEE
Binders, calcitriol, cinacalcet

WHAT IT ENCODES
Manage CKD-MBD with phosphate binders (sevelamer/non-calcium preferred), active vitamin D (calcitriol/paricalcitol), and calcimimetics (cinacalcet lowers PTH via CaSR).

BOARD TRAP
Cinacalcet LOWERS Ca (activates CaSR) — useful in secondary/tertiary hyperPTH; do NOT give in primary hyperPTH unless surgery is not an option, and watch for hypocalcemia.

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7. VDDR-1: ↓CYP27B1 — calcitriol responds

WHAT YOU SEE
Genetic rickets ghost, low 1,25

WHAT IT ENCODES
Vitamin D-dependent rickets type 1 (VDDR-1): loss of renal CYP27B1 (1-alpha-hydroxylase) → LOW 1,25-D with normal 25-D. Responds to physiologic calcitriol replacement.

BOARD TRAP
VDDR-1 has LOW 1,25 and corrects with calcitriol; VDDR-2 has HIGH 1,25 and does NOT (receptor defect) — the response to calcitriol distinguishes them.

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8. VDDR-2: VDR defect + alopecia

WHAT YOU SEE
Rickets ghost with alopecia

WHAT IT ENCODES
Vitamin D-dependent rickets type 2 (VDDR-2): vitamin D RECEPTOR (VDR) mutation → end-organ resistance → HIGH 1,25-D, often with ALOPECIA. Calcitriol-resistant; needs high-dose Ca (sometimes IV).

BOARD TRAP
Alopecia + rickets + sky-high 1,25-D that does NOT respond to calcitriol = receptor resistance (VDDR-2), not a synthesis defect.

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9. Acute pancreatitis — saponification

WHAT YOU SEE

Critical-care ward, pancreas

WHAT IT ENCODES

Acute pancreatitis causes hypocalcemia via saponification — free fatty acids from fat necrosis chelate calcium in the abdomen. Hypocalcemia is a poor-prognosis (Ranson) marker.

BOARD TRAP

Don't reflexively replete to normal in pancreatitis-related hypocalcemia unless symptomatic/QT prolonged — it often resolves; aggressive Ca can worsen pancreatic injury.

10. Rhabdo / tumor lysis / crush

WHAT YOU SEE

Crush-injury figure

WHAT IT ENCODES

Severe hyperphosphatemia from rhabdomyolysis, tumor lysis, or crush injury precipitates calcium-phosphate → hypocalcemia. High phosphate + low Ca + high K/urate suggests TLS.

BOARD TRAP

In rhabdo/TLS, AVOID aggressive calcium repletion (give only if symptomatic) — Ca-phosphate precipitation worsens AKI and nephrocalcinosis; treat the phosphate.

11. Citrate (blood/FFP) — chelation

WHAT YOU SEE

Blood/FFP bag, chelation

WHAT IT ENCODES

Massive transfusion or citrate anticoagulation (CRRT, apheresis) chelates ionized calcium → acute hypocalcemia despite normal total Ca. Monitor IONIZED Ca.

BOARD TRAP

Total Ca may read normal during citrate load — only IONIZED Ca reveals the deficit; replete ionized Ca during massive transfusion/citrate CRRT.

12. Hungry bone post-parathyroidectomy

WHAT YOU SEE

Bone avidly taking up Ca

WHAT IT ENCODES

Hungry bone syndrome: after parathyroidectomy for severe hyperPTH, demineralized bone avidly takes up Ca/PO4/Mg → profound prolonged hypocalcemia with LOW phosphate and Mg.

BOARD TRAP

Hungry bone = low Ca + low phosphate (bone uptake), whereas surgical hypoPTH = low Ca + HIGH phosphate — phosphate direction separates them post-op.

13. Mg–Ca–PO4 depletion triangle

WHAT YOU SEE

Red warning triangle Mg/Ca/PO4

WHAT IT ENCODES

The repletion trap: severe hypomagnesemia causes functional hypoPTH and PTH resistance → refractory hypocalcemia. Must correct Mg, Ca, AND phosphate together — fixing Ca alone fails if Mg is low.

BOARD TRAP

Giving Ca without repleting Mg in a hypomagnesemic patient (PPI/alcohol/diarrhea) will NOT correct the hypocalcemia — always check and replace all three.

14. Treatment ladder — acute to chronic

WHAT YOU SEE

Pharmacy ladder of repletion

WHAT IT ENCODES

Acute symptomatic: IV calcium gluconate (1–2 g 10% over 10–20 min) then infusion. Chronic: oral calcium + calcitriol; correct underlying cause (vitamin D, Mg, phosphate).

BOARD TRAP

Match the cause to the fix: vitamin D deficiency → ergocalciferol/cholecalciferol; CKD/hypoPTH → ACTIVE calcitriol; refractory → check Mg before escalating Ca.